Prediction of Bilingual Proficiency through Eye-Tracking Analysis of Mandarin Chinese-English Code-Switching: A Machine Learning Approach

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Background

- This study explores how the machine learning analysis of eye-tracking data can reveal the relationship between code-switching and visual attention in bilinguals, and ultimately, bilingual language proficiency.
- Studying language processing has revealed distinct eye movement patterns in bilinguals reading in their second language (L2), including longer reading times, more fixations, and shorter saccades compared to monolinguals, all of which reflect increased cognitive load (Mou et al., 2024).
- Code-switched sentences can predict bilingual language proficiency, offering a deeper understanding of bilingual cognitive control during reading.
- Eye-tracking technology provides valuable metrics, such as gaze duration and fixation count, which can reveal these cognitive processes.

Research Goals

- Analyze eye-tracking metrics from bilinguals reading code-switched Chinese-English sentences and utilize a machine learning-based Random Forest classification model.
- Develop a predictive model linking eye movement patterns to L2 language proficiency, offering new insights into bilingual cognitive processes.

Methods

1. **Dataset:** Adapted from *An Eye-Movement Database of Bilingual Language Control for Chinese-English Bilinguals* (Wang et al., 2023)
   - 80 Mandarin-English bilingual undergraduates completed the **Oxford Placement Test (OPT,** objective) and **Bilingual Switching Questionnaire** (BSWQ, self-report) to assess L2 proficiency prior to eye-tracking (*Table 2*)
   - Eye-movement metrics (while reading) included Regression Path Duration, Fixation Count, Dwell Time, etc.
2. **Data Preprocessing and Feature Engineering:**
   - Handled missing values (null filtering, **median/mean imputation**)
   - Removed outliers by **IQR filtering**
   - Applied **robust scaling** to reduce skew
   - Engineered features: temporal metrics, eg. regression-dwell ratio and fixation density
   - Data split **60/40** (train/test), **stratified** sampling
3. **Model Training & Evaluation:**
   - Trained **Random Forest Classifier** with top 5 eye-movement features
   - Hyperparameters tuned via **Randomized Search Cross-Validation**
   - Evaluation metrics: accuracy, weighted F1, ROC-AUC, confusion matrix, CV scores, and overfitting checks

Types	Language	Definition	Examples
Non-switch	Chinese	All words are Chinese.	一到教室,那些调皮的学生立马安静了下来。
	English	All words are English.	Upon entering the classroom, those naughty students quieted down immediately.
Insertion	Chinese	The subject noun of the main clause is in Chinese, while the rest of the sentence is in English.	Upon entering the classroom, those naughty 学生 quieted down immediately.
	English	The subject noun of the main clause is in English, while the rest of the sentence is in Chinese.	一到教室,那些调皮的students立马安静了下来。
Alternation	Chinese	The part of a sentence before the subject of the main clause is in English, and the remaining part is in Chinese.	Upon entering the classroom, those naughty 学生立马安静了下来。
	English	The part of a sentence before the subject of the main clause is in Chinese, and the remaining part is in English.	一到教室,那些调皮的students quieted down immediately.
Dense code-switching	Chinese	The first noun in the subordinate clause and the subject noun of the main clause are in Chinese, while all other words are in English.	Upon entering the 教室, those naughty 学生 quieted down immediately.
	English	The first noun in the subordinate clause and the subject noun of the main clause are in English, while all other words are in Chinese.	一到classroom,那些调皮的students立马安静了下来。

Table 1: *Illustrates examples of each of the 8 Code-Switching sentence conditions used in the existing eye-tracking study involving Mandarin-English bilinguals. All code-switched sentences retain the same meanings as their non-switched counterparts.*

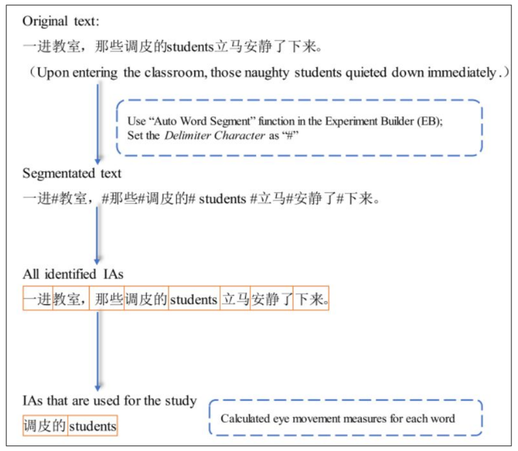


Figure 1: *The figure illustrates word segmentation during reading, with "#" inserted between words following Chinese usage patterns and between individual English words. Segmentation units (e.g., 教室 [jiào shì, "classroom"]) defined meaningful regions of interest for eye-tracking analysis.*

		High-proficient		Low-proficient	
		M	***SD***	***M***	***SD***
L1	Listening	4.475	0.554	4.452	0.629
	Speaking	4.225	0.620	4.448	0.581
	Reading	4.225	0.620	4.373	0.567
	Writing	3.700	0.564	3.893	0.527
	AoA	0.525	0.506	0.575	0.501
L2	Listening	3.025	0.620	2.288	0.659
	Speaking	3.100	0.632	1.998	0.627
	Reading	3.675	0.474	2.998	0.535
	Writing	3.100	0.379	2.355	0.588
	AoA	8.000	1.051	8.350	1.220
	OPT	52.500	2.736	31.280	3.863

Table 2: *Results of self-report questionnaire, OPT and BSWQ language evaluation tests of 80 bilingual participants (Wang et al., 2023)*

Results

Preprocessing improved data quality, reduced null values by **8%**, identified **6860 outliers** (27%, *Figure S6*). Feature selection revealed **top 5 most important predictors** with highest correlation: Dwell Time, Regression Path Duration, Fixation Count, First Fixation Duration and Fixation Density (*Figure 2*). Visualized features showed scaled feature distributions after data cleaning and filtering (*Figure S4*).

Class	Precision	Recall	F1-score	Support
High	0.67	0.75	0.71	16
Low	0.71	0.62	0.67	16
Accuracy			0.69	32
Macro Average	0.69	0.69	0.69	32
Weighted Average	0.69	0.69	0.69	32

Table 3: *Model Results of Random Forests Classifier after Hyperparameter Optimization*

We trained a Random Forest classifier on the **top five eye-movement features** to predict English L2 proficiency. Hyperparameters were optimized via **randomized search cross-validation** to reduce overfitting, with the **best RF configuration** including 50 trees, max depth of 5, min samples split of 10, min samples leaf of 8, and max features of 0.5, while applying class balancing (*Figure S9*). The model achieved an **69% accuracy**, a **macro-averaged F1-score of 0.69,** and a **test AUC of 0.81** *(Table 3).* The classifier showed a **stronger recall for high proficiency (H) learners** (0.75), while **precision was higher for low proficiency (L) learners** (0.71), suggesting the model was **more cautious but accurate** in detecting lower proficiency. **Training and test accuracy** (0.85 vs. 0.70) indicated **mild overfitting** but the model maintained **good generalization overall.**

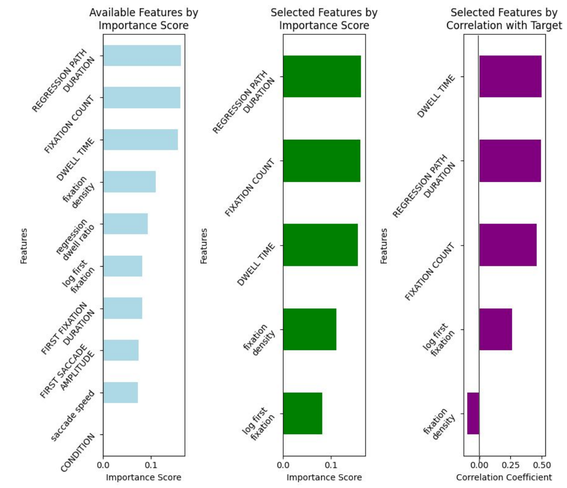


Figure 2: *Feature importance analysis showed that both late (Dwell Time, Regression Path Duration) and early (Fixation Count) eye-tracking measures were strong predictors of L2 proficiency, highlighting their roles in different stages of lexical processing.*

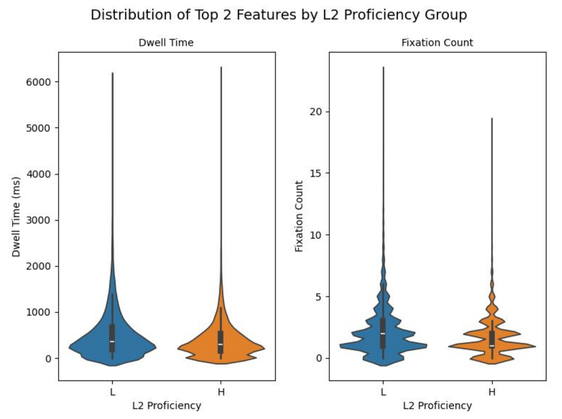


Figure 3: *Readers with lower L2 proficiency show longer dwell times and more fixations, indicating greater cognitive effort. Higher proficiency is associated with shorter, more efficient reading patterns.*

Discussion

This study provides empirical support that Mandarin-English bilinguals with lower L2 proficiency exhibit increased late-stage eye movement activity—such as longer fixation counts and regression path durations—indicating greater cognitive effort during reading. These findings align with prior research linking these metrics to challenges in comprehension and integration during L2 processing (Tywoniw, 2023).

Limitations include the use of pre-existing data, which restricts control over sentence design and participant variability. The binary proficiency classification oversimplifies bilingualism complexity, and combining self-reported and objective measures may introduce noise. While Random Forests performed well, their interpretability remains limited.

Future investigation will involve collecting original eye-tracking data using controlled stimuli and continuous proficiency scores. Employing interpretable models will help clarify how specific eye-movement patterns relate to language control and bilingual processing.

References

Supplemental Figures